

6.3

C Axis

The C Axis provides high-precision, bi-directional spindle motion that is fully interpolated with X and/or Z motion. You can command spindle speeds from 0.01 to 60 RPM.

C-Axis operation is dependent on the mass, diameter and length of the workpiece and/or the workholding (chuck). Contact the Haas Applications Department if any unusually heavy, large diameter, or long configuration is used.

6.3.1 Cartesian to Polar Transformation (G112)

Cartesian to Polar coordinate programming converts X,Y position commands into rotary C-Axis and linear X-axis moves. Cartesian to Polar coordinate programming greatly reduces the amount of code required to command complex moves. Normally a straight line would require many points to define the path, however, in Cartesian, only end points are necessary. This feature allows face machining programming in the Cartesian coordinate system.

C-Axis Programming Notes

Programmed moves should always position the tool centerline.

Tool paths should never cross the spindle centerline. If necessary re-orient the program so the cut does not go over the center of the part. Cuts that must cross spindle center can be accomplished with two parallel passes on either side of spindle center.

Cartesian to Polar conversion is a modal command. Refer to page **257** for more information on modal G-codes.

The G112 code is intended to be used with a Lathe using the C-axis and Live Tooling to program cutter anywhere along a non-rotating part.

The G112 code allows 3-D contouring using the X, Y, and Z axes. The tool centerline programming (G40) and cutter diameter compensation (G41/G42) are available with G112. They are also available for a tool in any of the three plane selection (G17, G18, G19).

A Lathe with Y-axis can use G112 and it can be useful to extend the range of travel of the live tool all the way across a part.

Circular motion (G02 and G03) in any of the three planes (G17, G18, G19) are also available with G112.

Since the spindle is not turning in G112, “feed per inch” (G98) must be selected.

Once G112 is active, all motions are programmed with XYZ and C cannot be used.

All X values are in radius when using G112.

Example Program

```
o51120 (CARTESIAN TO POLAR INTERPOLATION) ;
(G54 X0 Y0 is at the center of rotation);
(Z0 is on face of the part) ;
(T1 is an end mill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G20 G40 G80 G97 G99 (Safe startup) ;
G17 (Call XY plane) ;
G98 (Feed per min) ;
M154 (Engage C-Axis) ;
P1500 M133 (Live tool CW at 1500 RPM) ;
G00 G54 X2.35 C0. Z0.1 (Rapid to 1st position) ;
G112 (XY to XC interpretation);
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G0 X-.75 Y.5 ;
G01 Z0 F10.;
G01 X0.45 (Point 1) ;
G02 X0.5 Y0.45 R0.05 (Point 2) ;
G01 Y-0.45 (Point 3) ;
G02 X0.45 Y-0.5 R0.05 (Point 4) ;
G01 X-0.45 (Point 5) ;
G02 X-0.5 Y-0.45 R0.05 (Point 6) ;
G01 Y0.45 (Point 7) ;
G02 X-0.45 Y0.5 R0.05 (Point 8) ;
G01 X0.45 Y.6 (Point 9) ;
G00 Z0.1 (Rapid retract);
(BEGIN COMPLETION BLOCKS) ;
G113 (Cancel G112) ;
M155 (Disengage C axis) ;
M135 (Live tool off) ;
G18 (Return to XZ plane) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 (Z home) ;
M30 (End program) ;
```

6.3.2 Cartesian Interpolation

Cartesian coordinate commands are interpreted into movements of the linear axis (turret movements) and spindle movements (rotation of the workpiece).

Operation (M codes and Settings)

M154 engages the C-Axis and M155 disengages the C-Axis.

When not using G112, Setting 102 - Diameter is used to calculate the feed rate.

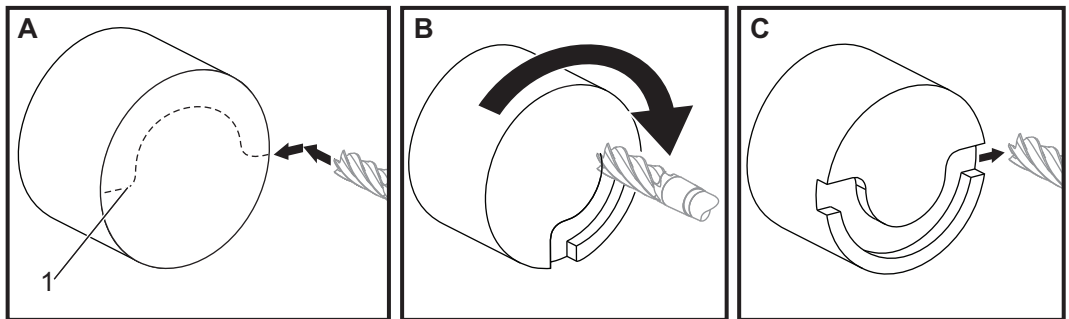
The lathe automatically disengages the spindle brake when you command the C-Axis to move and to reengage it afterwards if the M codes are still active.

C-Axis incremental moves are possible using the H address code as shown in this example:

```
G0 C90. (C-Axis moves to 90. deg.) ;  
H-10. (C-Axis moves to 80. deg. from the previous 90 deg  
position) ;
```

Sample Programs

F6.1: Cartesian Interpolation Example 1. (1) Projected Cutting Path (A) The endmill feeds 1" into the workpiece on one side. (B) The C Axis turns 180 degrees to cut the arc shape. (C) The endmill feeds 1" out of the workpiece.



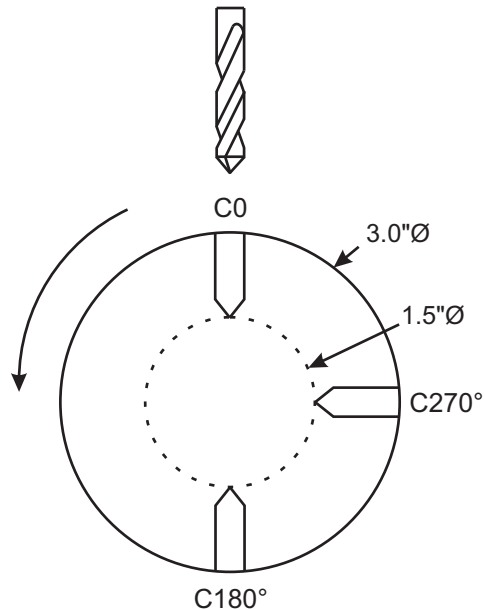
```
o51121 (CARTESIAN INTERPOLATION EX 1) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on face of the part) ;  
(T1 is an end mill) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G98 (Feed per min) ;  
M154 (Engage C Axis) ;  
G00 G54 X2. C90 Z0.1 (Rapid to 1st position) ;  
P1500 M133 (Live tool CW at 1500 RPM) ;  
M08 (Coolant on) ;  
(BEGIN CUTTING BLOCKS) ;  
G01 Z-0.1 F6.0 (Feed to Z depth) ;  
X1.0 (Feed to Position 2) ;
```

```

C180. F10.0 (Rotate to cut arc) ;
X2.0 (Feed back to Position 1 ) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z0.5 M09 (Rapid retract, coolant off) ;
M155 (Disengage C axis) ;
M135 (Live tool off) ;
G18 (Return to XZ plane) ;
G53 X0 Y0 (X & Y home) ;
G53 Z0 (Z home) ;
M30 (End program) ;

```

F6.2: Cartesian Interpolation Example 2



```

o51122 (CARTESIAN INTERPOLATION EX 2);
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on face of the part) ;
(T1 is a drill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G19 (Call YZ plane) ;
G98 (Feed per min) ;
M154 (Engage C-Axis) ;
G00 G54 X3.25 C0. Y0. Z0.25 ;
(Rapid to 1st position) ;
P1500 M133 (Live tool CW at 1500 RPM) ;

```